NumPy Array Slicing & Skewness

1. 1D Array Slicing

Given the following 1D array:

```
import numpy as np
arr = np.array([10, 20, 30, 40, 50, 60, 70, 80, 90])
```

1. Extract elements from index 2 to 6 (exclusive of 6).
2. Extract every second element from the entire array.
3. Reverse the entire array using slicing.
4. Extract the last 4 elements.
5. Extract elements from index 1 to 7 with a step of 2.

2. 2D Array Slicing

Given the following 2D array:

```
arr_2d = np.array([
  [1,  2,  3,  4,  5],
  [6,  7,  8,  9, 10],
  [11, 12, 13, 14, 15],
  [16, 17, 18, 19, 20]
])
```

1. Extract the entire second row.
2. Extract the entire third column.
3. Extract every alternate column from all rows.
4. Extract the bottom-right 2×2 sub-array.

3. 3D Array Slicing

Given the following 3D array:

```
arr_3d = np.arange(1, 25).reshape(2, 3, 4)
# Shape: (2, 3, 4)
```

1. Extract the second row from every 2D slice.
2. Extract the last column from every 2D slice.
3. Extract a sub-array of shape (2, 2, 2) from the top-left corner of each 2D slice.
4. Extract elements of all corners only 0 row and last row

4. Skewness Calculation of Raw Data

Dataset:

```
data1 = [12, 15, 14, 10, 18, 20, 22, 25, 13, 17, 19, 21, 16, 14, 23]
```

```
data2 =[5, 7, 8, 9, 10, 12, 15, 18, 22, 30, 45, 60]
```